

Content: Lectures 1-15

Lectures 1-7

1. Ordered Fields

- Examples/Non-examples
- Definition and properties of $\leq$

2. The Real Line

- Definition and properties of the absolute value
- Definition and properties of sup/inf
- The completeness axiom
- Archimedean property
- Dense sets

3. Metric Spaces

- Examples of metric spaces.
- Norms induce metrics
- Open/Closed sets in metric spaces
- Bounded sets.
- Interior, Closure, and Boundaries
- Equivalent norms and metrics

4. Sequences in Metric Spaces

- Bounded and convergent subsequences
- Subsequences

5. Sequences in $\mathbb{R}$

- Properties of convergent subsequences
- Monotone sequences
- Definition and properties of limsup/liminf

6. Completed Metric Spaces

- Cauchy sequences
- Definition of a complete metric space
- $\mathbb{R}$ is complete
- $\mathbb{R}^d$ is complete

7. Topological Spaces

- Definition of a Topological Space $(X, \mathcal{T})$.
- Metric Topology
- Convergence in Topological spaces
- Definition compact sets
- Closed $\cap$ compact is closed
- Nested sequences of compact sets have non-empty compact intersection.

8. Compact Sets in Metric Spaces.

- Definition of Sequentially compact
- Definition of Totally Bounded
- examples and non-examples of seq
- $\text{Compact} \implies \text{Seq. Compact} \implies \text{Totally Bounded and Complete} \implies \text{Compact}$.
- Heine-Borel Theorem.

9. Continuous functions

- Definition and equivalent definitions
- Cluster points
- Images of compact sets are compact
- Properties of real-values continuous functions (e.g. $f + g$ is continuous.)

10. Uniform continuity.

- Definition
- Examples/Non-examples
- For uniformly continuous f , x_n is Cauchy $\implies f(x_n)$ is Cauchy.
- Continuous functions with compact domain are uniformly continuous.

11. Sequences of functions and types of convergence

- Definition of Point-wise convergence
- Example where the limit of sequence of functions is not continuous.
- Definition of uniform convergence
- Example of uniformly converging sequence of functions.
- Uniform Cauchy Criterion
- For complete co-domains, Uniformly Cauchy $\iff$ Uniform convergence

12. Connected sets

- Definitions
- $[0, 1]$ is connected.
- Definition of path connected.
- Images of connected sets are connected.
- Examples of path connected and connected sets.
- Path connected implies connected. Converse is not true.
- Characterization of connected sets in $\mathbb{R}$.
- Open and connected in $\mathbb{R}^n$ implies path connected.
- The topologist's curve. (Do NOT need to know **how** to show it is connected but not path connected)

Practice Questions

- For each question, say whether such an object exists. If it does, give an example. If it does not, give a short explanation on why not. Assume all the sequences are in $\mathbb{R}$.
 - A sequence which is not monotone and has no convergent subsequence.
 - A Cauchy sequence with a divergent subsequence.
 - A sequence with exactly two subsequential limits.
 - A bounded sequence with no Cauchy subsequence

2. True or False.

- If (x_n) and (y_n) diverge, then the sequence $x_n y_n$ diverges.
- If $(x_n) \rightarrow 0$ and (y_n) is bounded then $x_n y_n \rightarrow 0$.
- Arbitrarily unions of closed sets are closed.
- Closed and bounded sets are compact in metric spaces.
- Let S be a nonempty bounded subset of $\mathbb{R}$ then $\inf S < \sup S$.

3. Prove the lower triangle inequality using the triangle inequality.

4. Use the fact that $\mathbb{Q}$ is dense to prove that there are infinitely many rationals in $[a, b]$ for $a, b \in \mathbb{R}$.

5. Assuming that $\mathbb{Q}$ is dense show that for all $x \in \mathbb{R}$ there exists an n such that $n \leq x \leq n + 1$.

6. Let X be a set of real numbers that has a supremum, and let $a := \sup X$. Prove that $X \cap (a - \varepsilon, a] \neq \emptyset$ for every $\varepsilon > 0$.

7. Prove that

$$\limsup_{n \rightarrow \infty} (s_n + t_n) \leq \limsup_{n \rightarrow \infty} s_n + \limsup_{n \rightarrow \infty} t_n$$

8. Suppose that a_n and b_n are Cauchy sequences. Prove that $(a_n b_n)$ is a Cauchy sequence without using the fact that Cauchy sequences converge.

9. Let A and B be subsets of a metric space E . Prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$|f(x) - f(y)| \leq |x - y|^2.$$

Prove that f is continuous, but not uniformly continuous.

11. Prove that for a bounded real sequence s_n that if $s_{n_k} \rightarrow L$ as $k \rightarrow \infty$ then

$$\liminf s_n \leq L \leq \limsup s_n.$$

12. Prove that for a real sequence s_n , $s_n \rightarrow \infty \iff s_n^{-1} \rightarrow 0$.

13. Let s_n be a sequence in a Metric space (E, d) . Prove that if $\forall n$

$$d(s_n, s_{n+1}) \leq d(s_n, s_{n-1})$$

is Cauchy.

14. Prove that $\lim n^{1/n} \rightarrow 1$.

15. Find the limit of

$$s_n = \frac{n^3 + 6n^2 + 7}{4n^3 + 3n - 4}$$

without using theorems from Calculus.

16. Prove that in a metric space all compact sets are totally bounded.
17. Let $f : X \rightarrow Y$ be a function such that for all compact $K \in Y$, $f^{-1}(K)$ is compact in X . Prove that f is continuous.
18. Which one of the following sets is compact in $\mathbb{R}$? Provide a brief explanation.
 - (a) $(0, 1]$
 - (b) $\{\frac{1}{n} : n \in \mathbb{N}\}$
 - (c) $[1, \infty)$
19. True or False.
 - (a) In a Metric space, Closed and bounded sets are compact.
 - (b) $\mathbb{Q}$ is connected.
 - (c) All compact sets are disconnected.
 - (d) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $A \subset \mathbb{R}$ is bounded then $f(A)$ is bounded.
 - (e) If A and B are disconnected then $A \cap B$ is disconnected.
20. Let $\{f_n : (E, d) \rightarrow (E', d')\}$ be a sequence of uniformly continuous functions that converge uniformly to f . Prove that f is uniformly continuous.
21. Let $\{f_n : (E, d) \rightarrow (E', d')\}$ be a sequence of bounded functions that converge uniformly to f . Prove that f is bounded.
22. Prove that $x^3 = \cos(x)$ for some x in $(0, \pi/2)$.
23. Suppose that $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at c and $g(c) \neq \hat{0}$ then there exists an open set U containing c such that $g(x) \neq 0$ for all $x \in U$.
24. Let X be a totally bounded space and $f : X \rightarrow \mathbb{R}$ be uniformly continuous. Prove that $f(X)$ is totally bounded.
25. Give an example of sequence of functions that converge uniformly and prove that is uniformly convergent.